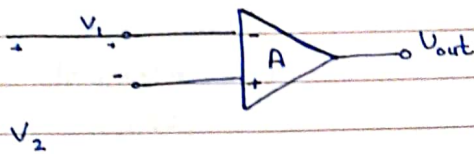


LINEAR

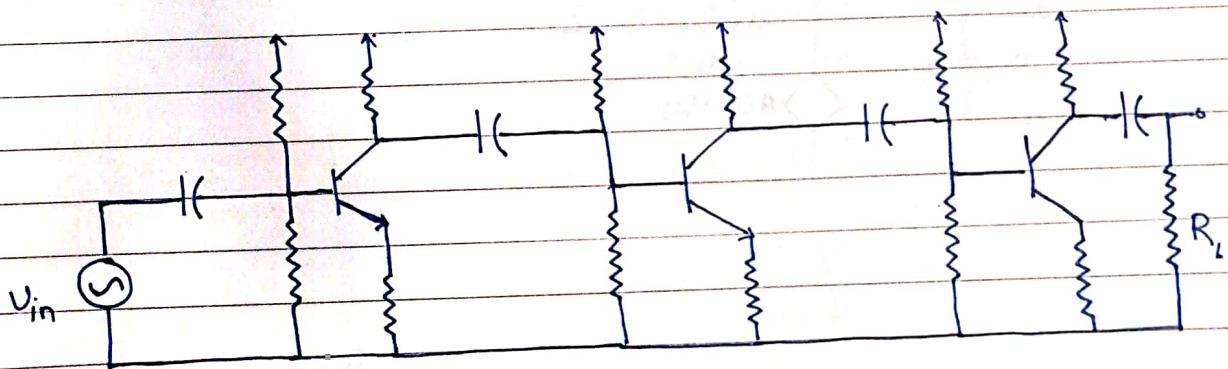
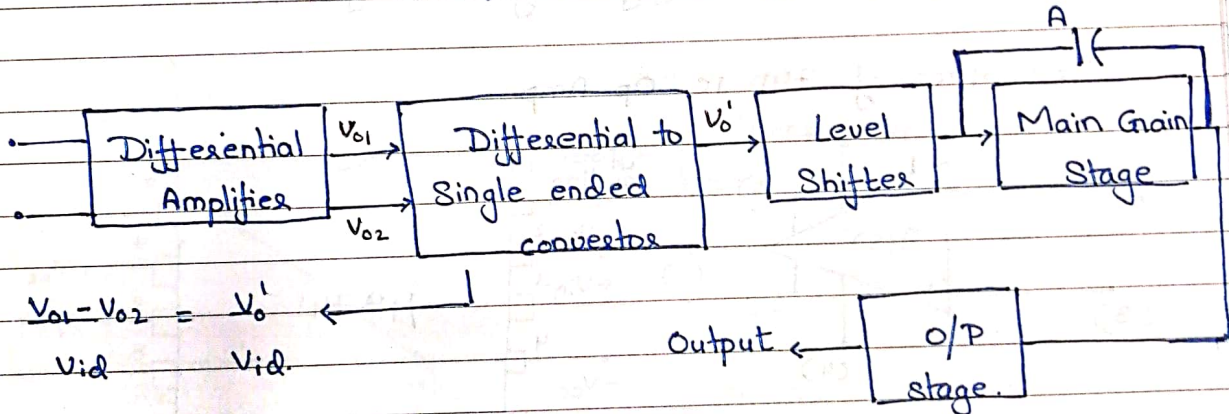
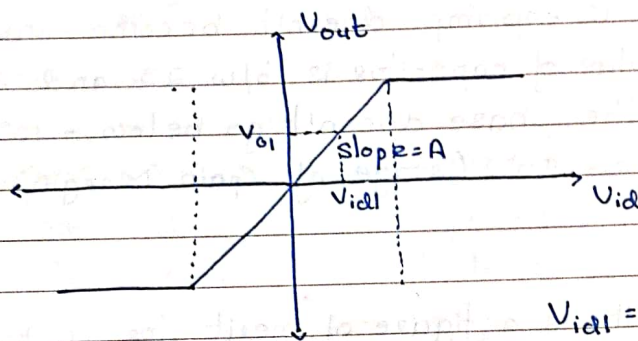
INTEGRATED

CIRCUITS.

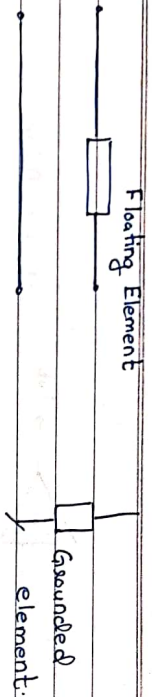


$$V_{out} = A(V_1 - V_2)$$

$$V_1 - V_2 = \frac{V_{out}}{A}$$



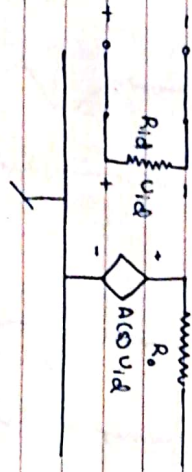
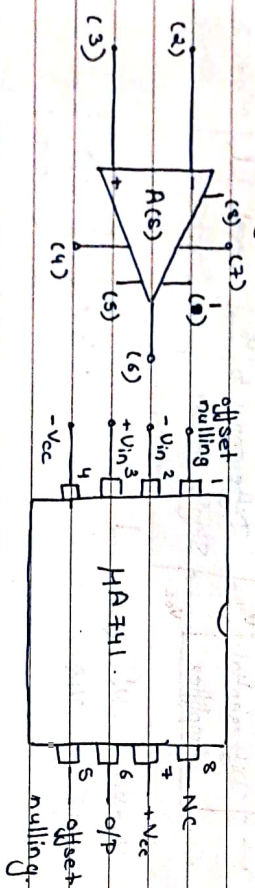
Sample IC. (Typical Voltage Amplifier Architecture)



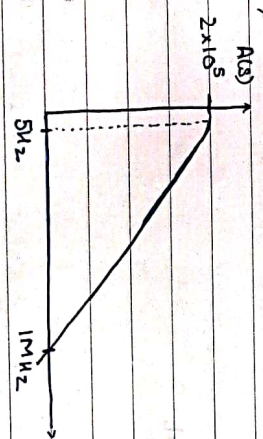
The compensating capacitor is provided in the IC Amp. introduces a dominant pole in the gain-frequency response and ensures that the IC-opamp doesn't become unstable since for IC 741 the value of capacitor is b/w 25 and 30 pf. This is because the phase doesn't go below -90° while phase can go as low to 180° (stage of gain Margin) which causes stability.

* The unity bandwidth is a figure of merit for IC 741.

Parameters of 741 IC Op-Amp.



	Ideal Value	Real Values.
Open loop gain	∞	$10^5 - 10^6$ (100-100dB)
Input Imped.	∞	3-5 M Ω
Output Imped.	0	50-75 Ω
Unity Gain Bandwidth	∞	1 MHz
Slew Rate	∞	0.5 V/ μ s
Input offset voltage	0	10-20 mV
Input-bias current	0	100-200 pA
Input offset current	0	40-60 pA
3dB Bandwidth of gain	∞	4-5 Hz



Slew rate is defined as the maximum value of rate of change of output voltage.

$\Rightarrow \frac{dV_o}{dt} \Big|_{\max}$

Typical values of slew rate for high speed op-amps. is 3000-10000 V/ μ s.

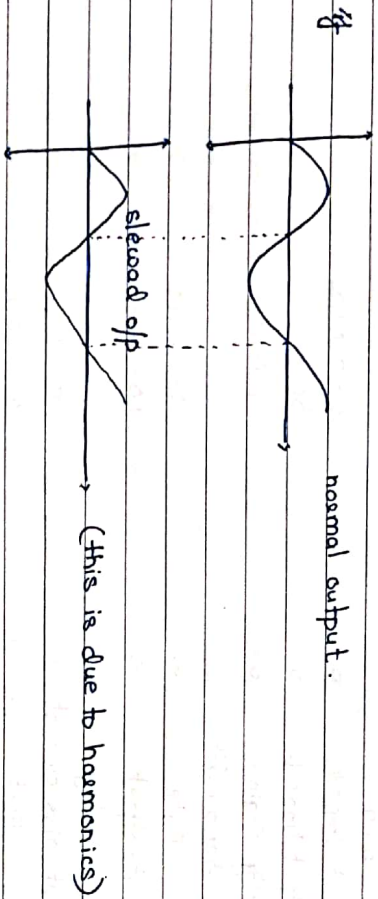


$$\frac{dV_o}{dt} = K V_i \omega \cos wt \Big|_{\max}$$

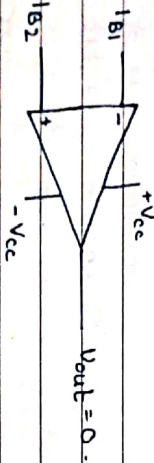
$= K \omega V_i \Rightarrow (\text{const})$
(fixed by IC manufacturer)

if $K=1$.

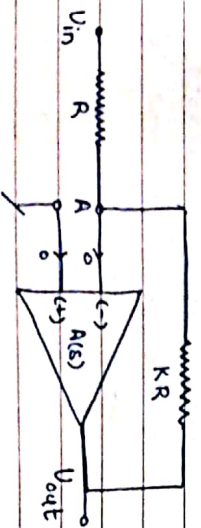
then $V_{ix} \omega = \text{const.}$



$(I_{B1} + I_{B2})/2 \Rightarrow$ Input bias current
 $(I_{B1} - I_{B2})/2 \Rightarrow$ Input offset current.



Inverting Amplifier



Assuming to be a non-ideal op-amp
 let voltage at A be V
 so K_{CL}
 $V_{in} - V = V - V_{out}$
 $R \quad KR$

and $A(s)(-V) = V_{out}$.

$$V_{in} = -V \left(\frac{K+1}{KR} \right) - \frac{V_{out}}{KR}$$

$$V_{in} = -V_{out} \left[\frac{(K+1)}{KR} + \frac{1}{K} \right]$$

$$V_{out} = \frac{-KA}{K+A+1} \Rightarrow \frac{-KA(s)}{A(s) + (K+1)}$$

$$\Rightarrow \frac{-K}{1 + (1+K)A(s)}$$

$\Rightarrow$ if $A(s) \rightarrow \infty$ (ideal case)
 then gain = $-K$. (closed loop gain)

this means in ideal case:

$$V_o = (A(s))(-V)$$

$$\Rightarrow V = 0$$

(hence pt. A is virtually grounded)
 (This is created by virtue of -ve feedback)

then in that case.

$$\frac{V_i - 0}{R} + \frac{V_o - 0}{KR} = 0$$

$$\frac{V_o}{V_i} = -K. \quad (\text{directly - for ideal case})$$

Different Models of the open-loop gain $A(s)$

1. Ideal op-amp model i.e. $A(s) \rightarrow \infty$

then $V_o = -K \cdot V_i$

2. $A(s) = \text{const. value} = A$.

let $K = 10$

$$\Rightarrow \frac{V_o}{V_i} = \frac{-10}{1 + 11/A} \approx \frac{-10A}{A+11}$$

$$\Rightarrow \text{if } A = 100$$

$$\approx \frac{-10}{1.11} = -9.009$$

if $A = 1000$

$$V_o = \frac{-10}{1.011} = -9.890$$

$$V_i = 1.011$$

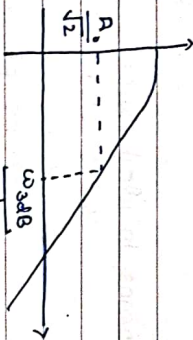
if $A = 10,000$

$$\frac{V_o}{V_i} = \frac{-10}{1.001} = -9.9989$$

3. $A(s) = \frac{A_o}{1 + s/\omega_{3dB}}$

$$1 + s/\omega_{3dB}$$

if $\omega \gg \omega_{3dB}$



then $A(s) \approx \frac{A_o \omega_{3dB}}{s}$ $2\pi \times 4.5 \text{ rad/s}$

(since $S=j\omega$)

$$\Rightarrow A(s) = \frac{A_o}{1 + s/\omega_{3dB}} = \frac{A_o \omega_{3dB}}{s + \omega_{3dB}} \quad (1\text{-pole model of open loop})$$

$\Rightarrow A(s) = \frac{\omega_t}{s}$ (where $\omega_t = A_o \omega_{3dB}$) $\rightarrow$ Integral model of $A(s)$ function.

Dependence of Closed Loop Gain (Transfer Function) on frequency:

$$\text{Let } A(s) = \frac{\omega_t}{s}$$

$$T(s) = \frac{V_o/V_i}{1 + (1+K)} = \frac{-K}{1 + (1+K)} \quad \frac{\omega_t/s}{\omega_t/s}$$

$$= \frac{-K \times \omega_t}{1+K}$$

$$\frac{\omega_t}{1+K} + s$$

$\Rightarrow$ single pole mode.

3dB bandwidth of closed loop gain

$$= \frac{\omega_t}{1+K}$$

DC closed loop gain $= \frac{K}{1+K}$ (mod-value)
product of gain and bandwidth.

$$\Rightarrow K \omega_t = \omega_t \quad \text{if } K \gg 1$$

i.e. constant.

Open loop Gain Closed Loop Gain

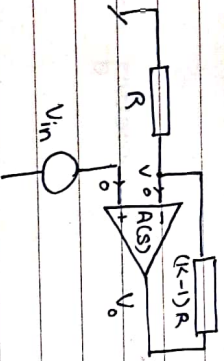
DC-gain A_o

3dB-bandwidth ω_{3dB}

$$G_{CL} B \cdot \omega_t \quad A_o \cdot \omega_{3dB} = \omega_t$$

$$\frac{K \omega_t}{1+K} \approx \omega_t \quad (K \gg 1)$$

Non-Inverting Amplifier.



$$\frac{V_o - V_i}{R} + \frac{V_o - V_i}{(K-1)R} = 0$$

$$V_o = \frac{V_i}{K-1} \left(\frac{K-1+1}{K-1} \right)$$

$$= \frac{V_i}{K-1} \left(\frac{K}{K-1} \right)$$

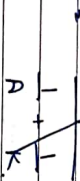
$$\text{and } V_o = (V_{in} - V_i) \cdot A$$

$$V = V_{in} - \frac{V_o}{A}$$

$$V_o (1 + \frac{K}{A}) = \frac{V_{in} K}{A(K-1)}$$

$$V_o (A(K-1) + K) = V_{in} K$$

$$\frac{V_o}{V_{in}} = \frac{K A}{A(K-1) + K}$$



$$V_o = K (V_{in} - V_o)$$

$$V_o (1 + K) = K V_{in}$$

$$\frac{V_o}{V_{in}} = \frac{K}{1 + K} \Rightarrow \frac{1}{1 + \frac{1}{K} A(s)}$$

$\Rightarrow$ if $A(s) = \omega_t / s$

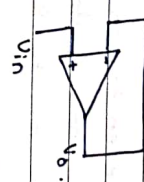
$$\frac{1}{\frac{s}{\omega_t} + 1} = \frac{\omega_t}{s + \omega_t}$$

closed loop B.W. = ω_t / K

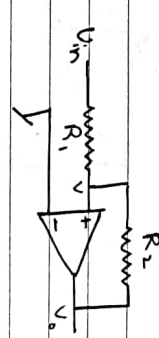
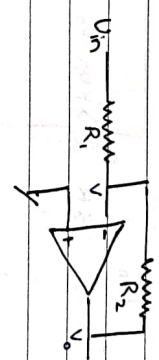
DC gain = K

G.B.P (Gain Bandwidth Product) = ω_t (non-inverting amp.)

In order to achieve unity gain non-inverting amp. (with open loop gain: inf)



& Find the expression of closed loop gain of these cases by taking $A(s) = \omega_t / s$



(done-before)

$$V_{in} - V + V_o - V = 0$$

$$V_o = A(s) V$$

$$V_{in} = \frac{-V_o + V}{R_1 R_2} \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

$$V_{in} = \frac{+V_o (-\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_1})}{R_1 R_2 A(s)}$$

$$V_{in} = \frac{V_o (-A(s) R_1 + R_2 + R_1)}{A(s) R_1 R_2}$$

$$V_o = \frac{-A R_2}{A R_1 (R_1 + R_2)} V_{in}$$

$$= \frac{-A}{A R_1 (R_1 + R_2)}$$

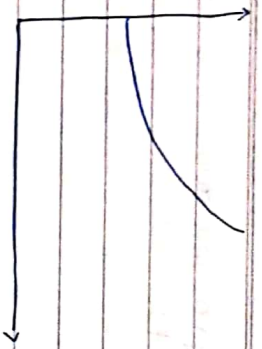
$$= \frac{-\omega_t R_1 / s}{(1 + R_1 / R_2)}$$

$$= \frac{-\omega_t R_1}{s R_2} - (1 + R_1 / R_2)$$

$$(K = R_2 / R_1)$$

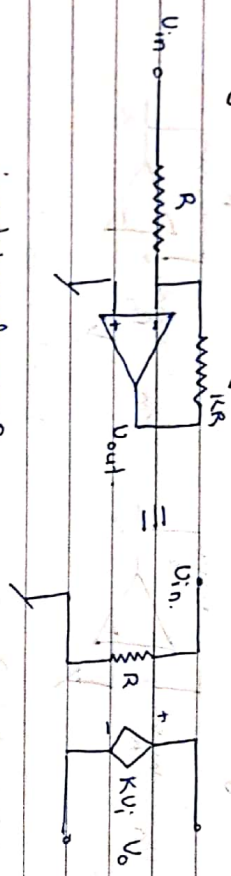
$$= \frac{-\omega_t (1 + R_1 / R_2)}{s - \omega_t (R_1 / R_2 + R_1)}$$

$$\Rightarrow \omega_t (K+1) / s, \omega_t (K+1)$$

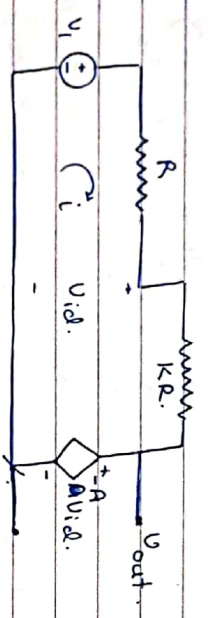


(since pole is in R.H. of s-plane)
even though for an impulse input output saturates to $-R_2/R_1$.

Realization of different types of Controlled sources.
(1) Voltage Controlled Voltage Source.



input impedance = R .
output " = 0 .



$$V_1 - i(k+1)R + AV_{id} = 0$$

$$\text{and } V_1 - iR = V_{id}$$

$$\text{and } V_0 = -AV_{id}$$

$$\Rightarrow V_1 - i(k+1)R + A(V_1 - iR) = 0$$

$$(A+1)V_1 = i(k+1+AR)$$

$$\Rightarrow i = \frac{A+1}{k+1+AR} V_1$$

$$V_1 \left(\frac{A+1}{k+1+AR} \right) = iR$$

$$\Rightarrow V_1 \left(\frac{k+1}{k+1+AR} \right) = V_{id}$$

$$i = \frac{A+1}{A+(k+1)} \cdot \frac{V_1}{R}$$

$$\Rightarrow V_1 - \frac{A+1}{A+(k+1)} \cdot V_1 = V_{id}$$

$$\Rightarrow V_1 \left(\frac{k}{A+(k+1)} \right) = V_{id}$$

$$\text{and } V_0 = -AV_{id}$$

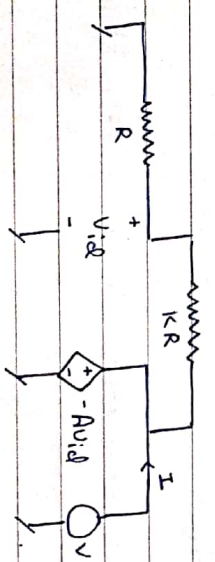
$$= -\frac{kA}{A+(k+1)} V_1$$

$$= \frac{-k}{1+(k+1)/A} V_1$$

$$\text{for } A \rightarrow \infty$$

$$\Rightarrow V_0 = -kV_1$$

for output impedance.



$$\text{since } AV_{id} = V_{id}$$

$$\Rightarrow V_{id} = 0$$

$$\text{so } I \rightarrow \infty$$

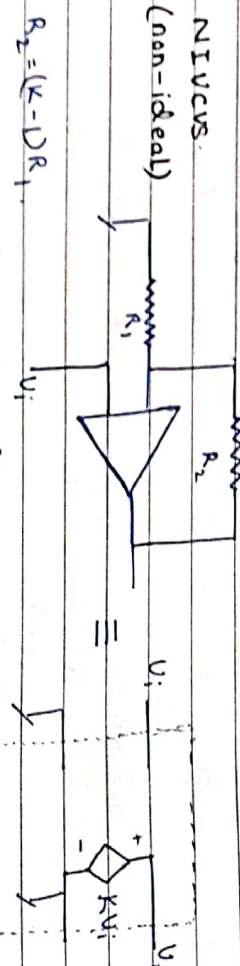
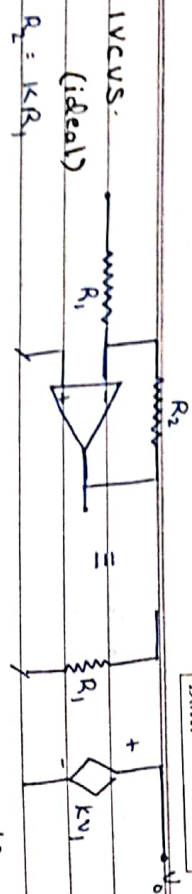
$$\text{and } R_0 = V/I = 0$$

$$i = \frac{V_1}{1+(k+1)/A}$$

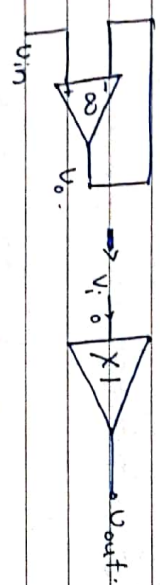
$$\text{for } A \rightarrow \infty$$

$$i = \frac{V_1}{R}$$

(input impedance)

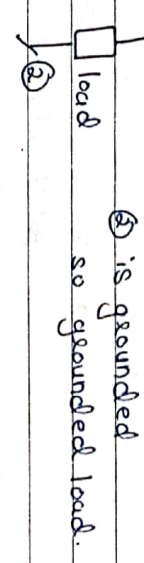
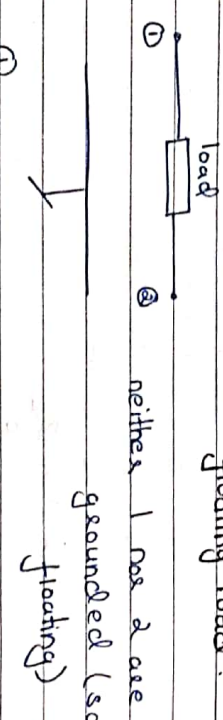
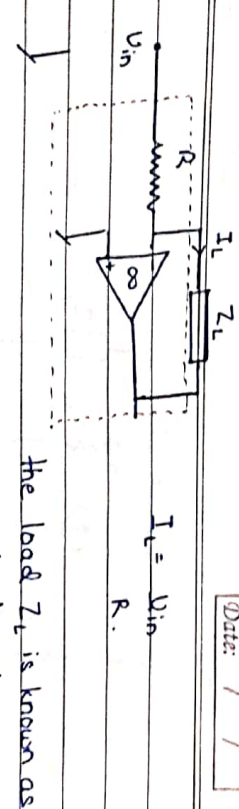
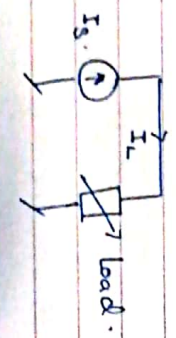


Special: Voltage Buffers.

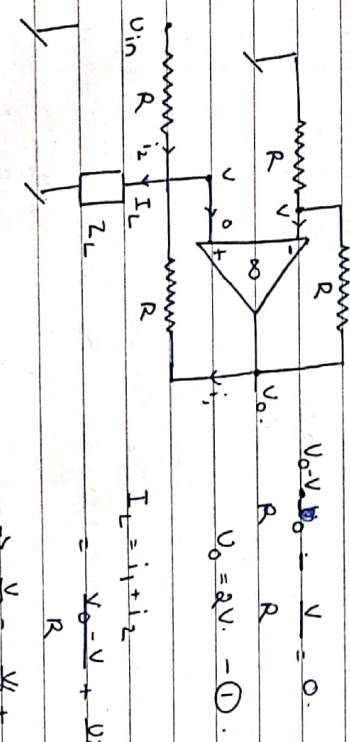


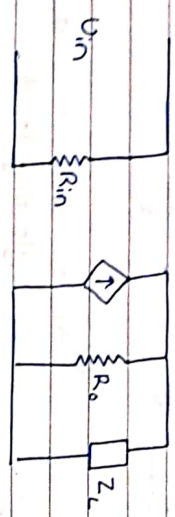
Note: never write KCL at output node. Use it only at input terminals.

a) Voltage Controlled Current Source.



It is necessary to have a floating load to realize a VCCS.





$$I_L = \frac{V_{in}}{R} = \frac{V_o}{2Z_L}$$

$$\Rightarrow V_o = 2V_{in} \cdot Z_L$$

So $V_{in} - i_2 R = V_o / 2$
 $V_{in} - i_2 R = 2V_{in} \cdot Z_L$

$$V_{in} (1 - 2Z_L) = i_2 R$$

$$R_{in} = \frac{V_{in}}{i_2} = \frac{R}{(1 - 2Z_L)} \quad (\text{input impedance})$$

for R_o

$$i_{out} = i_1 + i_2$$

$$i_1 = \frac{V_o - V}{R} \Rightarrow \frac{V}{R} - \frac{V_o}{R} = \frac{V_o}{2R}$$

$$i_2 = \frac{V_o - V}{2R} \text{ or } \frac{V_o}{2R}$$

$$\Rightarrow \frac{V_o}{R + (R \parallel Z_L)}$$

$$\Rightarrow i_{out} = V_o \left(\frac{1}{2R} + \frac{1}{R + (R \parallel Z_L)} \right)$$

$$i_o = \frac{V_o}{\left(\frac{1}{2R} + \frac{1}{R + (R \parallel Z_L)} \right)}$$

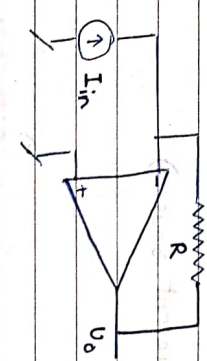
$$\Rightarrow \frac{2R}{R(R + Z_L) + RZ_L}$$

$$\Rightarrow \frac{2R}{R^2 + 2RZ_L + 2R^2 + 2RZ_L}$$

$$\Rightarrow \frac{3R^2 + 4RZ_L}{2R(R^2 + 2RZ_L)}$$

So $R_o = \frac{2R(R^2 + 2RZ_L)}{3R^2 + 4RZ_L}$

Current Controlled Voltage Source.



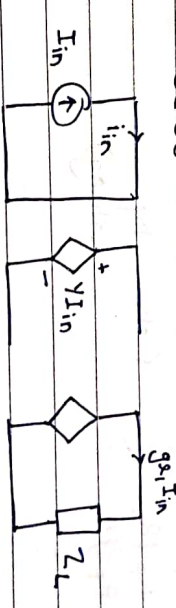
$$V_o + I_{in} R = 0 \Rightarrow V_o = -I_{in} R$$

$R_{in} = 0$ (since $V = 0$ by virtual ground)
 (current dependent source)

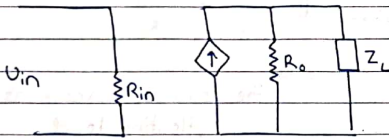
$R_o = 0$ ($\because V_o = 0$ by virtual ground)
 ideal.

Current Controlled Current Source.

$$CCCS = \beta V_{in} \text{ VCCS}$$



$V_2 = V_1$



$$I_L = \frac{V_{in}}{R} = \frac{V_o}{2Z_L}$$

$$\Rightarrow V_o = 2V_{in} - Z_L$$

$$\text{So } V_{in} = i_2 R = V_o / 2$$

$$V_{in} = i_2 R = 2V_{in} Z_L / R$$

$$V_{in} (1 - 2Z_L / R) = i_2 R$$

$$R_{in} = \frac{V_{in}}{i_2} = \frac{R}{(1 - 2Z_L / R)} \quad (\text{input impedance})$$

for R_o

$$i_{out} = i_1 + i_2$$

$$i_1 = \frac{V_o - V}{R} \Rightarrow \frac{V}{R} - \frac{V}{2R} = \frac{V_o}{2R}$$

$$i_2 = \frac{V_o - V}{R} \Rightarrow \frac{V_o - 0}{R} = \frac{V_o}{R + (R \parallel Z_L)}$$

$$\Rightarrow \frac{V_o}{R + (R \parallel Z_L)}$$

$$\Rightarrow i_{out} = V_o \left(\frac{1}{2R} + \frac{1}{R + (R \parallel Z_L)} \right)$$

$$\frac{V_o}{i_o} = \frac{1}{\left(\frac{1}{2R} + \frac{1}{R + (R \parallel Z_L)} \right)}$$

$$\Rightarrow \frac{1}{\frac{2R}{R + Z_L} + \frac{R + Z_L}{R(R + Z_L) + RZ_L}}$$

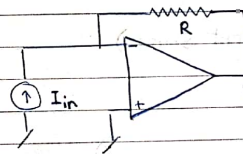
$$\Rightarrow \frac{1}{\frac{2R}{R^2 + 2RZ_L} + \frac{R + Z_L}{R^2 + 2RZ_L}}$$

$$\Rightarrow \frac{R^2 + 2RZ_L + 2R^2 + 2RZ_L}{2R(R^2 + 2RZ_L)}$$

$$\Rightarrow \frac{3R^2 + 4RZ_L}{2R(R^2 + 2RZ_L)}$$

$$\text{So } R_o = \frac{2R(R^2 + 2RZ_L)}{3R^2 + 4RZ_L}$$

Current Control Voltage Source.



$$V_o + I_{in} R = 0$$

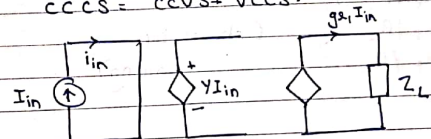
$$\Rightarrow V_o = -I_{in} R \quad (\text{current dependent source})$$

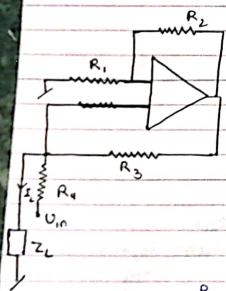
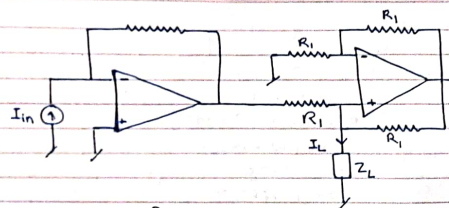
$$R_{in} = 0 \quad (\text{since } V = 0 \text{ by virtual ground})$$

$$R_o = 0 \quad (\because V_o = 0 \text{ by virtual ground ideal})$$

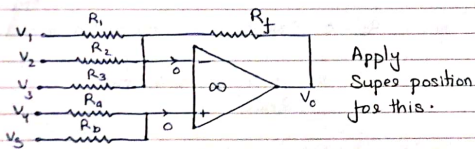
Current Controlled Current Source.

CCCS = CCVS + VCCS.





Determine the condition for which I_L is independent of load Z_L .
(Howland's Current Source)



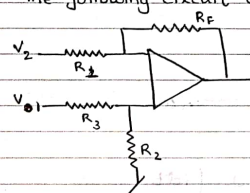
$$\begin{aligned} V_{out}(V_1) &= -R_F/R_1 \cdot V_1 \\ V_{out}(V_2) &= -R_F/R_2 \cdot V_2 \\ V_{out}(V_3) &= -R_F/R_3 \cdot V_3 \end{aligned} \quad \text{(check by solving)}$$

$$\begin{aligned} V_{out}(V_4) &= V_4 (R_b/R_a + R_b) (1 + (R_f/R_1 \parallel R_2 \parallel R_3)) \\ V_{out}(V_5) &= V_5 (R_a/R_a + R_b) (1 + (R_f/R_1 \parallel R_2 \parallel R_3)) \end{aligned}$$

$$\text{hence } V_{out} = K_5 V_5 + K_4 V_4 - (K_1 V_1 + K_2 V_2 + K_3 V_3)$$

where $K_1, \dots, K_5$ are const. developed due to resistors.

The following circuit can be called as weighted ~~summer~~ ^{summer}.



Under what conditions is the relationship b/w the diff. R's) $V_{out} = V_1 = V_2$.

$$\text{for } V_2 \quad V_{out} = -\frac{R_F}{R_1} V_2$$

Fig: Difference Amplifier.

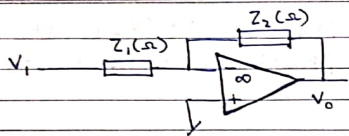
$$\text{for } V_1 \quad V_{out} = V_1 \cdot \left(\frac{R_F}{R_2 + R_3} \right) \left(1 + \frac{R_F}{R_1} \right)$$

$$R_1 = R_F - \textcircled{1} \quad \text{and} \quad R_2 = R_3 - \textcircled{2}$$

$$\Rightarrow V_{out} = V_1 \left(\frac{1}{1 + R_3/R_2} \right) \left(1 + \frac{R_F}{R_1} \right) - \frac{R_F}{R_1} V_2$$

$$\text{or } \frac{R_F}{R_1} = \frac{R_3}{R_2} = 1 \quad (\text{if not equal to } 1)$$

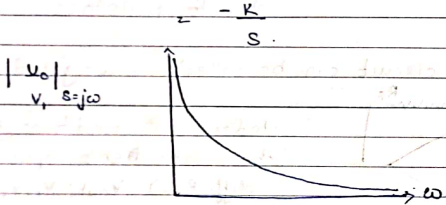
$$V_o = V_1 - \frac{R_F}{R_1} V_2$$



- $Z_2 =$ ① $1/cs$
 ② LS
 ③ $R+LS$
 ④ Rcs
 $1+Rcs$

$$\frac{V_o}{V_i} = -\frac{Z_2}{Z_1} = -\frac{(1/cs)}{R} \quad \text{if } Z_1 = R(s)$$

$$Z_2 = 1/cs(s)$$



if $I = V_i/R$

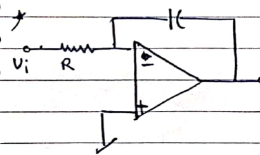
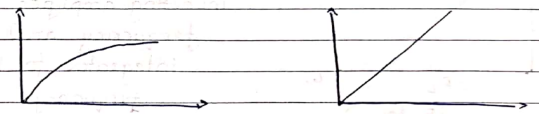
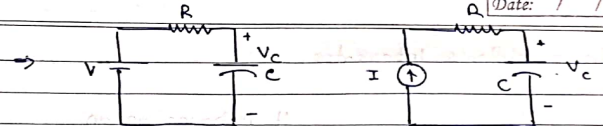
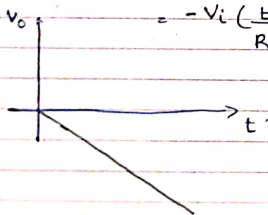
$$C \frac{dV_c}{dt} = \frac{V_i}{R} \quad \text{or} \quad V = \frac{1}{C} \int i dt$$

$$= \frac{1}{C} \int i dt \Rightarrow \frac{1}{C} \int \frac{V_i}{R} dt$$

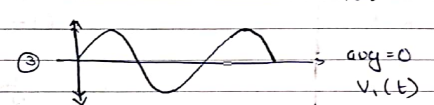
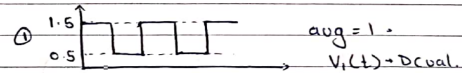
(linear)

$$V_c = \frac{1}{RC} \int V_i dt$$

$$V_o = -V_c = -\frac{1}{RC} \int V_i dt$$

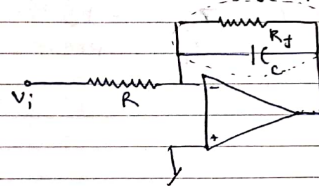


(Integrator ckt)



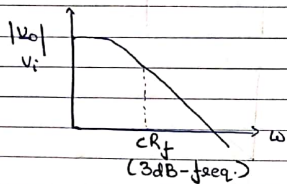
Only use when you are sure that there is no DC offset.
Since for DC val. capacitor becomes o.c and $V = V_{sat}$.

To solve this use R_f in parallel.



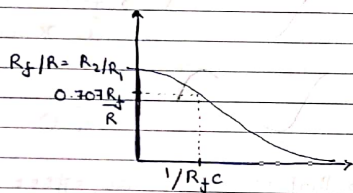
$$\begin{aligned}
 \frac{V_o}{V_i} &= -\frac{Z_2}{Z_1} \\
 &= -\frac{1}{R} \left(\frac{R_f / cs}{R_f + 1/cs} \right) \\
 &= -\frac{R_f}{R} \cdot \frac{1}{Rcs + 1}
 \end{aligned}$$

This is Miller's Integrator.



It behaves as an inverting amplifier for low frequency and integrator for high frequency.

$$\rightarrow \frac{V_o}{V_i} = - \frac{1}{CR} \left(\frac{1}{s + 1/R_f C} \right)$$

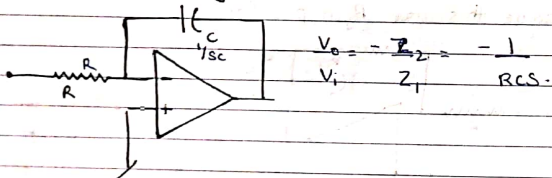


(low pass filter function)

for ideal integrator slope happens to be -20dB decade.

(- sign gives that it is an inverting amplifier)

Typical Miller's integrator.



let

$$(i) Z_i = R_i + 1/sC_i \equiv \frac{R_i(s) \cdot 1/sC_i(s)}{R_i(s) \cdot 1/sC_i(s)}$$

$$(ii) Y_i = G_i + sC_i = \frac{1/sC_i(s)}{R_i(s)}$$

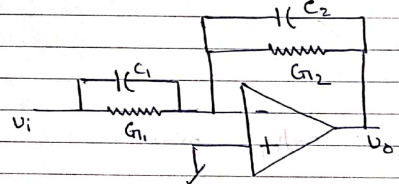
First Order - Bilinear T.F.

$$\text{if } Y_i = G_i + sC_i$$

$$Y_2 = G_2 + sC_2$$

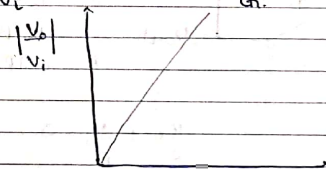
for Miller's integrator

$$\frac{V_o}{V_i} = - \frac{Y_1}{Y_2} = - \frac{G_1 + sC_1}{G_2 + sC_2} = - \frac{(s + G_1/C_1) C_1}{s + G_2/C_2 C_2}$$



if we consider a differentiator

$$\frac{V_o}{V_i} = - sCR = - sC$$



let T.F be $\frac{a_1 s + a_0}{b_1 s + b_0} = -Ks$

$$\left(\frac{a_1/a_0 s + 1}{b_1/b_0 s + 1} \right) \frac{b_0}{a_0}$$

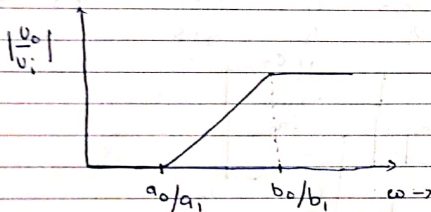
OR

$$\Rightarrow \omega_{n1} = a_0/a_1 \text{ (zero)}$$

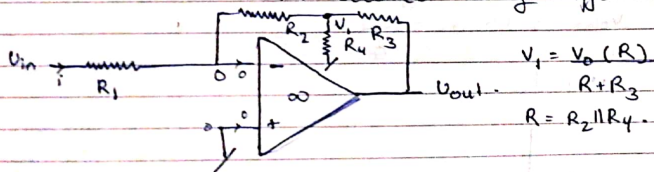
$$\omega_{n2} = b_0/b_1 \text{ (pole)}$$

for differentiator:

assume a_0/a_1 comes before
i.e. $a_0/a_1 < b_0/b_1$



for $\omega > a_0/a_1$ and $\omega < b_0/b_1$
T.F. becomes that of differentiator.



by virtual ground,

$$\frac{V_{in}}{R_1} + \frac{V_1}{R_2} = 0$$

$$\frac{V_1}{R_1} = -\frac{R_2}{R_1} \frac{V_{in}}{R_1}$$

$$\begin{aligned} \frac{V_o}{V_{in}} &= -\frac{R_2}{R_1} \left(\frac{R+R_3}{R} \right) \quad \text{for } R_4 \rightarrow \infty \\ &\Rightarrow -\frac{R_2}{R_1} \left(1 + \frac{R_3(R_2+R_4)}{R_2 R_4} \right) \\ &\Rightarrow -\frac{R_2}{R_1} \left(1 + \frac{R_3 + R_3}{R_4 R_2} \right) \end{aligned}$$

$$\begin{aligned} \text{for } R_{in} &\Rightarrow V_{in} - iR_1 = 0 \\ &\Rightarrow V_{in} = R_1 i = R_{in} i \end{aligned}$$

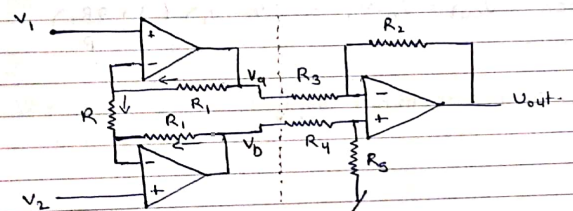
$$\begin{aligned} \text{if } R_{in} &= 5M\Omega \\ V_o &= -10^4 V_i \\ \text{and } R_2, R_3, R_4 &\leq 5M\Omega \end{aligned}$$

$$\begin{aligned} R_2 &= 5M\Omega, R_2 = 2 \times 10^6 \\ R_3 &= 9999\Omega, R_3 = 2.5 \times 10^6 \\ R_4 &= 1\Omega, R_4 = 100\Omega \end{aligned}$$

let $R_3 + R_4 = R_1$

$$\text{then } \frac{V_o}{V_i} = - \left(\frac{R_2}{R_4} + \frac{R_3}{R_1} \right)$$

$$\begin{aligned} \text{if } R_3 &= 5 \times 10^6 \text{ and } R_4 = 100\Omega \\ R_2 &= 1 \times 10^6 \Rightarrow 9.99 \times 10^4 \end{aligned}$$



$$V_{out} = -V_a \left(\frac{R_2}{R_3} \right) + V_b \left(\frac{1 + R_2/R_3}{1 + R_4/R_5} \right)$$

by virtual ground.
 $K_{OL} = 1$.

$$V_a = V_1 = \frac{V_1 - V_2}{R} \quad \text{--- (1) (current direction in diagram)}$$

$$K_{OL} = 2$$

$$\frac{V_b - V_2}{R_1} + \frac{V_1 - V_2}{R} = 0 \quad \text{--- (2) (assume both entering +ve)}$$

$$\Rightarrow V_a = \left(\frac{R_1 + 1}{R} \right) V_1 - \frac{R_1}{R} V_2$$

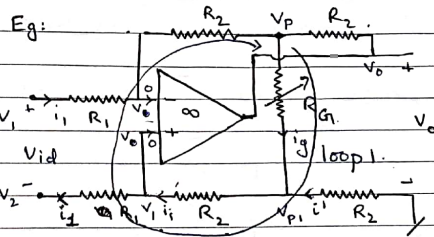
$$V_b = - \frac{R_1}{R} V_1 + \left(\frac{R_1 + 1}{R} \right) V_2$$

$$V_a - V_b = \left(\frac{2R_1 + 1}{R} \right) (V_1 - V_2)$$

$$\text{if } R_2 = R_3 \\ R_4 = R_5$$

$$\text{then } V_{out} = V_b - V_a = (V_2 - V_1) \left(\frac{1 + 2R_1}{R} \right)$$

This is instrumentation amplifier.



$$i_1 = \frac{V_{id}}{2R_1}$$

$$V_b - V_p = \frac{V_p - V}{R_2} + \frac{V_p - V_{p1}}{R_{G1}} \quad \text{--- (1)}$$

$$\frac{V_p - V}{R_2} + i_1 = 0 \Rightarrow \frac{V_p - V}{R_2} = -i_1 \quad \text{--- (2)}$$

$$i_1' = i_1 - i_g$$

$$i_1 = i_1' + i_g$$

$$i_1 = \frac{V_p - V_{p1}}{R_{G1}} + \frac{0 - V_{p1}}{R_2}$$

$$V_b = V_p - \left[i_1 - \frac{V_p - V_{p1}}{R_{G1}} \right] R_2$$

for loop 1.

$$V_p = V_{p1} - 2i_1 R_2$$

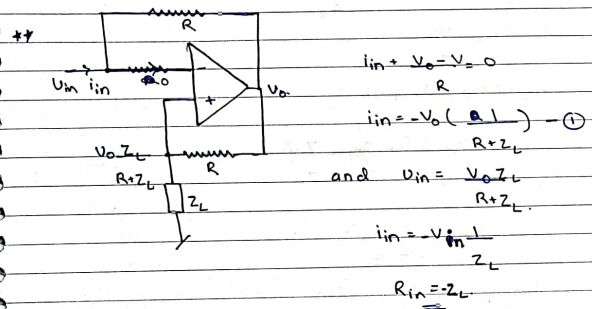
$$V_p - V_{p1} = -2i_1 R_2$$

at node V_p .

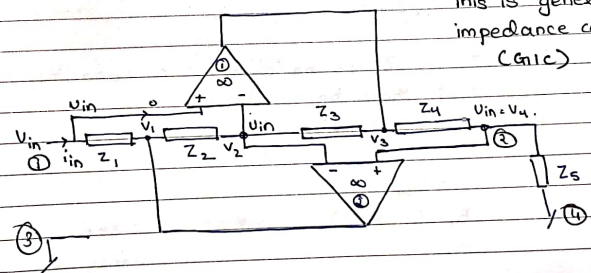
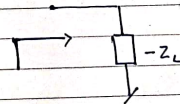
$$i_1 + \frac{V_p - V_{p1}}{R_{G1}} = \frac{V_p - V_b}{R_2}$$

$$i_1 - 2i_1 R_2 = \frac{V_p - V_b}{R_2}$$

Ideal inductor: ∞ Q (quality factor) $= \omega L/R \rightarrow \infty$
Real inductor: ∞ $Q = \frac{\omega L}{R}$



This circuit is known as a negative impedance converter or NIC. The equivalent ckt gives.



$$V_{in} - V_1 = I_{in} Z_1$$

and $V_1 - V_{in} = I_{in} Z_2$ $V_{in} - V_3 = I_{in} Z_3$

$$V_3 - V_{in} = I_{in} Z_4$$

$$\Rightarrow V_3 = V_{in} \left(1 + \frac{Z_4}{Z_3} \right)$$

$$\Rightarrow V_1 - V_{in} = I_{in} \left(1 - \frac{Z_4}{Z_3} \right)$$

$$\Rightarrow V_1 = V_{in} + I_{in} \left(\frac{-Z_4 Z_2}{Z_3 Z_5} \right)$$

$$\Rightarrow \frac{V_{in} - V_1}{Z_1} = I_{in} \left(1 - \frac{Z_4 Z_2}{Z_3 Z_5} \right) = I_{in}$$

$$V_1 - V_{in} = V_{in} \left(\frac{-Z_4 Z_2}{Z_3 Z_5} \right)$$

$$\Rightarrow I_{in} = + V_{in} \left(\frac{Z_2 Z_4}{Z_1 Z_3 Z_5} \right)$$

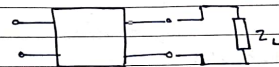
$$\Rightarrow R_{in} = \frac{Z_1 Z_3 Z_5}{Z_2 Z_4}$$

if either of Z_2 or Z_4 is taken as a capacitor, then (let others be resistance)

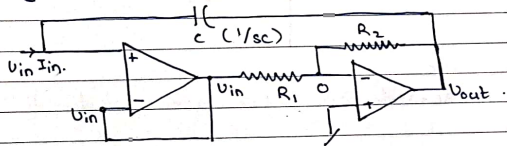
$$\Rightarrow R_{in} = \frac{R_1 R_2 R_3}{R_4} (sC)$$

$$Leq = \frac{R_1}{R_4} (R_2 R_3 C) \Rightarrow \frac{R^2 C}{s} \Rightarrow \omega \text{ sec} \Rightarrow \text{Henry (H)}$$

This forms a 2-port ckt.



Eg:



$$V_o - V_{in} + I_{in} = 0$$

$$\Rightarrow \frac{1}{sC} (V_{in} (R_2 + 1)) = I_{in} R_1$$

$$V_{in} + V_o = 0$$

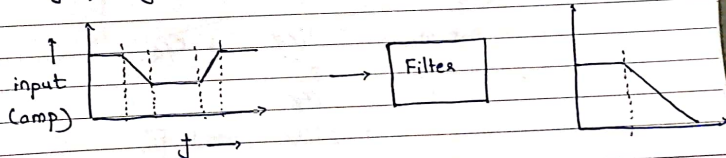
$$Z_{in} = \frac{V_{in}}{I_{in}} = \frac{1}{sC (1 + R_2/R_1)}$$

$$\Rightarrow V_o = \left(\frac{-R_2}{R_1} \right) V_{in} \quad \text{--- (1)}$$

$$C_{in} = \frac{C (1 + R_2/R_1)}{R_1}$$

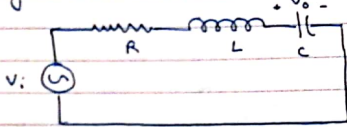
Filter

A frequency selective ckt.



Output wave may correspond to a different waveform of the figure.

let for a series R-L-C ckt.



$$\begin{aligned} \frac{V_o}{V_i} &= \frac{1/sC}{R + sL + 1/sC} \\ &\Rightarrow \frac{1}{RCS + s^2LC + 1} \\ &\Rightarrow \frac{1}{LCs^2 + RCS + 1} = \frac{1/LC}{s^2 + (R/L)s + 1/LC} \end{aligned}$$

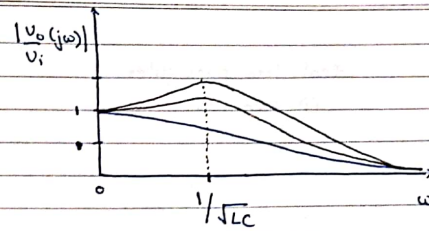
$$\begin{aligned} \Rightarrow \omega_n^2 &= 1/LC \\ \Rightarrow \omega_n &= 1/\sqrt{LC} \end{aligned}$$

for $s = j\omega \Rightarrow \frac{V_o(j\omega)}{V_i} \Rightarrow \frac{1/LC}{(1/LC - \omega^2) + j(R/L)\omega}$

magnitude $\Rightarrow \frac{1/LC}{\sqrt{(1/LC - \omega^2)^2 + (R/L\omega)^2}}$

at $\omega_n \Rightarrow \frac{1/LC}{R/L\omega_n} = K/R$

$$\frac{1/LC}{R/L\omega_n} = \frac{1}{R} \sqrt{\frac{L}{C}}$$



at $\omega \rightarrow \infty$

$$|V_o(j\omega)| = 0$$

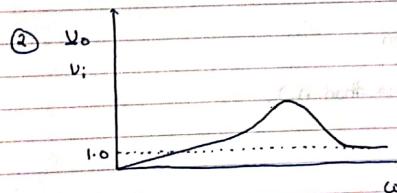
3 diff. waves.
 $\Rightarrow$ dependent upon different values of R, L and C.

if output is taken across inductor;

$$\begin{aligned} \frac{V_o}{V_i} &= \frac{sL}{R + sL + 1/sC} \\ &\Rightarrow \frac{s^2LC}{s^2LC + RCS + 1} \Rightarrow \frac{s^2}{s^2 + (R/L)s + 1/LC} \end{aligned}$$

in case output is across resistor.

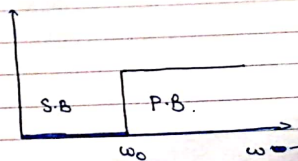
$$\begin{aligned} \frac{V_o}{V_i} &= \frac{R}{R + sL + 1/sC} \\ &\Rightarrow \frac{RCS}{s^2LC + RCS + 1} \Rightarrow \frac{(R/L)s}{s^2 + (R/L)s + 1/LC} \end{aligned}$$

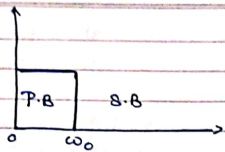


We are considering the primary criteria that the given T.F has poles in left half of s-plane.

ideal ~~ideal~~ high pass filter

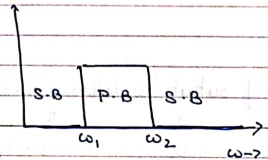
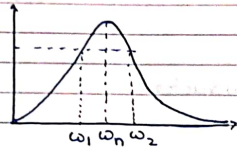
S.B: Stop Band P.B: Pass Band





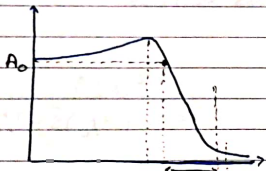
ideal low pass filter.
① = ideality.

③



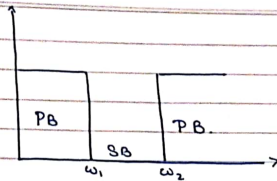
ideal bandpass filter.

The ideal cases are known as brick wall structure.



transition band.
(gain less than A_0)

④



Band reject / Band elimination / Notch filter.

gain for $\omega: \omega_1$ to ω_2 amplitude is 0.

⑤



allows all frequencies, but phase is different for different frequencies.

for ① T.F: $\frac{H \omega_0^2}{s^2 + s \frac{\omega_0}{Q} + \omega_0^2}$

② T.F: $\frac{H \cdot s^2}{s^2 + s \frac{\omega_0}{Q} + \omega_0^2}$

③ T.F: $\frac{H \cdot s (\omega_0/Q)}{s^2 + s (\omega_0/Q) + \omega_0^2}$

④ T.F: $\frac{s^2 + \omega_0^2}{s^2 + s (\omega_0/Q) + \omega_0^2}$

⑤ T.F: $\frac{s^2 - s (\omega_0/Q) + \omega_0^2}{s^2 + s (\omega_0/Q) + \omega_0^2}$